

Domains

Overall required score: 76%

Your score: 73%

Failed

Evidence Preservation

Your score: 83%

- Understand the methodology of collection 2/2
- Preserve and maintain the integrity of evidence 1/2
- Execute the steps of collection plan based on best practices 2/2

Storage Device Fundamentals

Your score: 50%

- Demonstrate how to analyze logical storage characteristics 0/2
- Demonstrate how to analyze physical device characteristics 3/4

Fundamentals of Digital Forensics

Your score: 60%

- Identify evidence of execution on Windows operating systems 2/2
- Understand the basic structure of a digital forensic report 1/2
- Analyze digital forensics artifacts on Windows operating systems 3/6

Digital forensic tools and techniques

Your score: 100%

- Demonstrate appropriate usage of log and timeline tools and techniques 3/3
- Demonstrate appropriate usage of network analysis tools and techniques 3/3
- Demonstrate appropriate usage of digital forensic analysis tools and techniques 2/2

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 1/30

Logical acquisition is taking a forensic copy of the whole _____.

☐ Folder (directory)

☐ System

☒ Partition/Volume

☐ Disk

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 2/30

When creating a file with a bit-stream copy of the disk, this is called:

- ☐ File to disk copy
- ☐ Image to disk
- ☒ Disk to image copy
- ☐ Disk to disk copy

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 3/30

Static or dynamic acquisition: Which is better?

- ☐ Dynamic, because you can get a copy of what is actually running on the system at that time
- ☐ None of them, actually network acquisitions must be performed first
- ☐ Static, because you can take a full forensic image without getting interrupted by a running process
- ☒ It just depends on the case you are dealing with

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 6/30

Validating your evidence could be done by:

- ☐ Storing evidence in a secure container
- ☐ Asking someone to double check your work after you
- ☒ Hashing Algorithms
- ☐ Signing a document

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 7/30

The actual entries of a CustomDestinations-ms JumpList are _____.

- ☐ Pointers to the actual files
- ☐ Normal URLs
- ☐ LNK files
- ☒ OLE streams

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 8/30

Whenever a major event occurs on a Windows operating system, such as a driver or update, it automatically generates a _____.



Restore Point



Backup of system files



Restore Location



Backup of important files

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 9/30

Assume you have a case where a malicious EXE was launched on a system. Which artifact would give you an idea of what files it used and called during its execution?



Registry files



Prefetch files



AppCompatCache files



LNK files

Questions summary



A progress bar for 'Questions summary' consisting of 15 circles. The first circle is orange, and the remaining 14 are grey.

Lab questions



A progress bar for 'Lab questions' consisting of 15 circles. The first circle is orange, and the remaining 14 are grey.

Q: 12/30

The _____ is the disk drive part that is concerned with reading from or writing on the _____.

- ☐ Head, spindle
- ☐ Platter, head
- ☒ Head, platter
- ☐ Head, cylinder

Q: 12/30

The _____ is the disk drive part that is concerned with reading from or writing on the _____.

- ☐ Head, spindle
- ☐ Platter, head
- ☒ Head, platter
- ☐ Head, cylinder

Q: 12/30

The _____ is the disk drive part that is concerned with reading from or writing on the _____.

- ☐ Head, spindle
- ☐ Platter, head
- ☒ Head, platter
- ☐ Head, cylinder

Q: 12/30

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- ☐ Head, spindle
- ☐ Platter, head
- ☒ Head, platter
- ☐ Head, cylinder

Q: 12/30

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- ☒ Head, platter
- ☐ Head, cylinder

Q: 12/30

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- ☐ Head, spindle
- ☐ Platter, head
- ☒ Head, platter
- ☐ Head, cylinder

Questions summary



A progress bar for 'Questions summary' consisting of 15 circles. The 2nd circle from the left is highlighted in blue, indicating the current question.

Lab questions



A progress bar for 'Lab questions' consisting of 15 circles. The 8th circle from the left is highlighted in white, indicating the current question.

- Pinned
- Answered
- Unanswered

Q: 13/30

Which of the following is not a file attribute?

- ☐ Hidden
- ☐ Archive
- ☐ Permissions
- ☒ Volume label

Q: 13/30

Which of the following is not a file attribute?

- ☐ Hidden
- ☐ Archive
- ☐ Permissions
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Q: 13/30

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Q: 13/30

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Q: 13/30

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Q: 13/30

Which of the following is not a file attribute?

- ☐ Hidden
- ☐ Archive
- ☐ Permissions
- ☒ Volume label

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 14/30

Which of the following protocols below generates GET/POST requests?



IMAP



HTTP



FTP



DNS

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 15/30

Syslog facilities are used to describe _____ .



The origin of log



The time when the log was generated



The size of log



The type of log

Questions summary



Lab questions



Pinned

Answered

Unanswered

Q: 16/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. How many files were found deleted in the recycle bin?

☒ 10

☐ 9

☐ 8

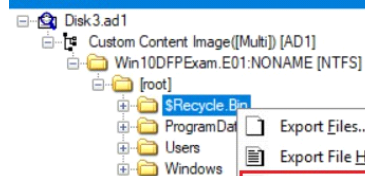
☐ 7

AccessData FTK Imager 4.7.1.2

File View Mode Help



Evidence Tree



File List

Name	Size	Type	Date Modified
S-1-5-18	1	Directory	3/13/2018 3:53:47 PM
S-1-5-21-319621592-4104958386-1817415...	1	Directory	3/13/2018 8:37:15 PM

- Export Files...
- Export File Hash List...
- Export Logical Image (AD1)...**
- Add to Custom Content Image (AD1)

Properties



Select Image Destination

×

Image Destination Folder

C:\Users\Administrator\Desktop\RecycleBin

Browse

Image Filename (Excluding Extension)

Disk3

Image Fragment Size (MB)

1500

For Raw, E01, and AFF formats: 0 = do not fragment

Compression (0=None, 1=Fastest, ..., 9=Smallest)

6

Use AD Encryption

☐

Filter by File Owner

☐

< Back

Finish

Cancel

Help

Create Image

×

Image Source

\$Recycle.Bin

Starting Evidence Number:

0

Image Destination(s)

C:\Users\Administrator\Desktop\RecycleBin\Disk3-1 [Logical image]

Add...

Edit...

Remove

Add Overflow Location

☒ Verify images after they are created

☐ Precalculate Progress Statistics

☐ Create directory listings of all files in the image after they are created

Start

Cancel

AccessData FTK Imager 4.7.1.2

- File

View

Mode

Help
- Add Evidence Item...

Add All Attached Devices

Image Mounting...

Remove Evidence Item

Remove All Evidence Items

Create Disk Image...

Export Disk Image...

Export Logical Image (AD1)...

Add to Custom Content Image (AD1)

Create Custom Content Image (AD1)...

File List

Name	Size	Type	Date Modified
------	------	------	---------------

Mount Image To Drive



Add Image

Image File:

C:\Users\Administrator\Desktop\RecycleBin\disk3.ad1



Mount Type: Logical Only



Drive Letter: Next Available (E:)



Mount Method: File System / Read Only



Write Cache Folder:

C:\Users\Administrator\Desktop\RecycleBin



Mount

Mapped Image List

Mapped Images:

Mount Image To Drive



Add Image

Image File:



Mount Type: Physical & Logical



Drive Letter: Next Available (E:)



Mount Method: Block Device / Read Only



Write Cache Folder:

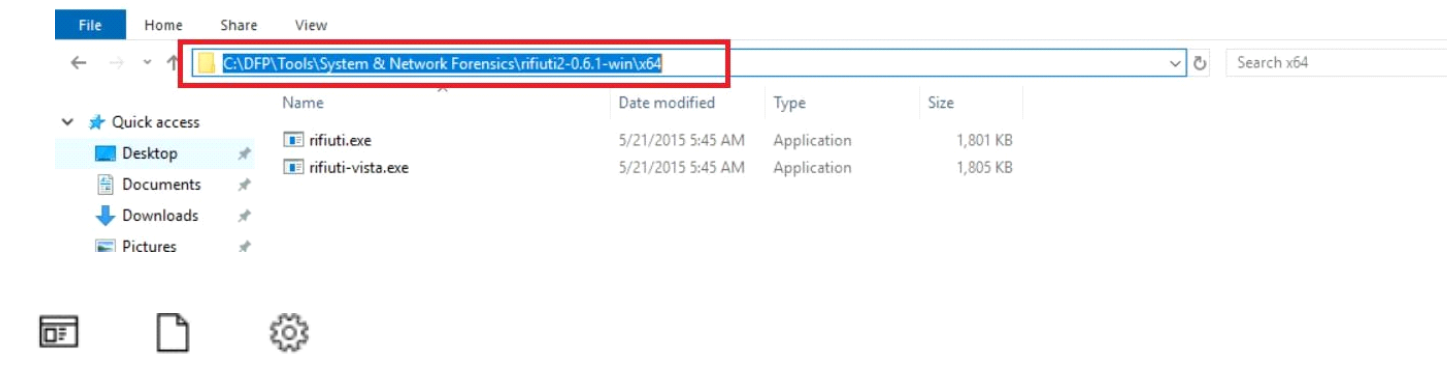
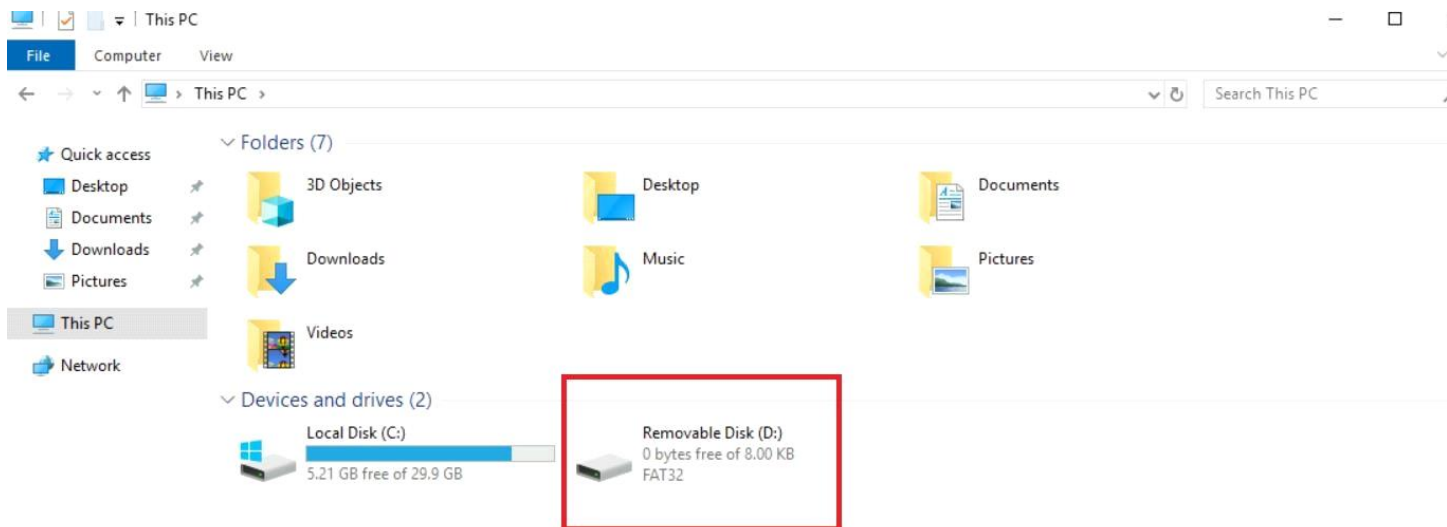


Mount

Mapped Image List

Mapped Images:

Drive	Method	Partition	Image
D:	File System/Read Only	File System	C:\Users\Administrator\Desktop\Recycle



Best match



Command Prompt

Desktop app

Apps



Log Parser 2

Settings (1)

Run as administrator

Run as different user

Open file location

Pin to Start

Pin to taskbar

C:\ Administrator: Command Prompt

```
C:\Users\Administrator>cd C:\DFP\Tools\  
C:\DFP\Tools>cd "System & Network Forensics\rifiuti2-0.6.1-win\x64"  
C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>
```

S-1-5-21-319621592-4104958386-1817415613-1001

File Home Share View

← → ↻ ↗

D:\S-1-5-21-319621592-4104958386-1817415613-1001

Search S-1-5-21-319621592-4...

Name	Date modified	Type	Size
SRGD3Y7R	3/13/2018 7:45 PM	File folder	
\$I0I2QTU.exe	3/13/2018 7:45 PM	Application	1 KB
\$I5YZIUG.xml	3/13/2018 7:42 PM	XML Document	1 KB
\$I30	3/13/2018 8:37 PM	File	4 KB
\$IB7O0LV	3/1/2018 5:49 AM	Shortcut	1 KB
\$ID2B777.xml	3/13/2018 8:37 PM	XML Document	1 KB
\$I0I2QTU.exe	3/13/2018 7:45 PM	Application	1 KB

Quick access

- Desktop
- Documents
- Downloads
- Pictures
- This PC

rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-1817415613-1001

C:\ Administrator: Command Prompt

```
C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-1817415613-1001  
Recycle bin path: 'D:\S-1-5-21-319621592-4104958386-1817415613-1001'  
Version: 2  
  
Index Deleted Time Size Path  
$IB7O0LV.lnk 2018-03-01 05:49:21 1107 C:\Users\dfp\Desktop\Quick Stego.lnk  
$I5YZIUG.xml 2018-03-13 19:42:26 153143 C:\Users\dfp\Documents\netscan.xml  
$ID1X4E3.docx 2018-03-13 19:42:44 13195 C:\Users\dfp\Documents\FronPublic\ExamQuestions.docx  
$I0870Y4.bmp 2018-03-13 19:45:17 3240054 C:\Users\dfp\Desktop\CyberAttack-Info.bmp  
$IQ0SPGZ.exe 2018-03-13 19:45:54 531368 C:\Users\dfp\Desktop\Utils\putty.exe  
$I0I2QTU.exe 2018-03-13 19:45:57 1541744 C:\Users\dfp\Desktop\Utils\splitme.exe  
$IKS6DLY.exe 2018-03-13 19:45:58 461952 C:\Users\dfp\Desktop\Utils\runtime.exe  
$IGD3Y7R 2018-03-13 20:34:57 359336 C:\Users\dfp\Desktop\Utils  
$IORULNX.lnk 2018-03-13 20:34:59 865 C:\Users\Public\Desktop\notepad++.lnk  
$ID2B777.xml 2018-03-13 20:37:15 1907 C:\Users\dfp\Documents\Nmap\network-scan.xml  
  
C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-1817415613-1001_
```


○ Unanswered

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. How many drive letters were attached to this system?

2

The screenshot displays the Registry Explorer application interface. On the left, the 'Key name' tree shows the path C:\Users\Administrator\Desktop\registry hives\SYSTEM expanded, with 'MountedDevices' selected. The right pane, titled 'Values', lists four registry values under the 'DosDevices' key:

Value Name	Value Type	Data
DosDevices[C:]	RegBinary	27-98-9C-7C-00-00-60-22-00-00-00-00
V??(Volume{e262f6ae-1d07-11e8-9531-806e5f6e5963})	RegBinary	5C-00-3F-00-3F-00-5C-00-53-00-43-00-00
DosDevices[D:]	RegBinary	5C-00-3F-00-3F-00-5C-00-53-00-43-00-00
V??(Volume{b26bd47c-1d0e-11e8-9534-0800dead128b})	RegBinary	5F-00-3F-00-3F-00-5F-00-55-00-53-00-00
DosDevices[E:]	RegBinary	5F-00-3F-00-3F-00-5F-00-55-00-53-00-00

Below the values list, the 'Type viewer' section shows a hex dump of the first value's data.

Type viewer
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 00000000 27 9B 9C 7C 00 00 60 22 00 00 00 00

Questions summary



Lab questions



● Pinned

● Answered

● Unanswered

Q: 18/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. What was the recycle bin's metadata file holding the runtime.exe file?

☐ \$IORVLNX.lnk

☒ \$IKS6DLY.exe

☐ \$IQOSPGZ.exe

☐ \$I0I2QTU.exe

Administrator: Command Prompt

C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-1817415613-1001

Recycle bin path: 'D:\S-1-5-21-319621592-4104958386-1817415613-1001'

Version: 2

Index	Deleted	Time	Size	Path
\$IB700LU.lnk	2018-03-01	05:49:21	1107	C:\Users\dfp\Desktop\Quick Stego.lnk
\$I5YZIUG.xml	2018-03-13	19:42:26	153143	C:\Users\dfp\Documents\netscan.xml
\$IDI84E3.docx	2018-03-13	19:42:44	13195	C:\Users\dfp\Documents\FromPublic\ExamQuestions.docx
\$I087AY8.bmp	2018-03-13	19:45:17	3240054	C:\Users\dfp\Desktop\CyberAttack-Info.bmp
\$IQOSPGZ.exe	2018-03-13	19:45:54	531368	C:\Users\dfp\Desktop\Utils\putty.exe
\$I0I2QTU.exe	2018-03-13	19:45:57	1541744	C:\Users\dfp\Desktop\Utils\splitme.exe
\$IKS6DLY.exe	2018-03-13	19:45:58	461952	C:\Users\dfp\Desktop\Utils\runtime.exe
\$IGD3Y7R	2018-03-13	20:34:57	359336	C:\Users\dfp\Desktop\Utils
\$IORVLNX.lnk	2018-03-13	20:34:59	865	C:\Users\Public\Desktop\notepad++.lnk
\$ID2B777.xml	2018-03-13	20:37:15	1907	C:\Users\dfp\Documents\Nmap\network-scan.xml

Questions summary



Lab questions



- Pinned
- Answered
- Unanswered

Q: 19/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. Using the given JumpLists, from where was the "ExamQuestions.pdf" accessed?



\\SHARES.DFP.EXAM\PUBLIC



\\SHARES.DFP.EXAM\



C:\Users\dfp\Documents

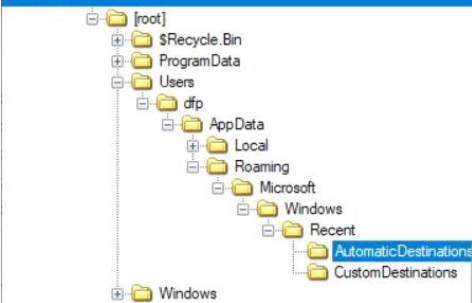


C:\Users\dfp\Documents\Exams

AccessData FTK Imager 4.7.1.2

File View Mode Help

Evidence Tree



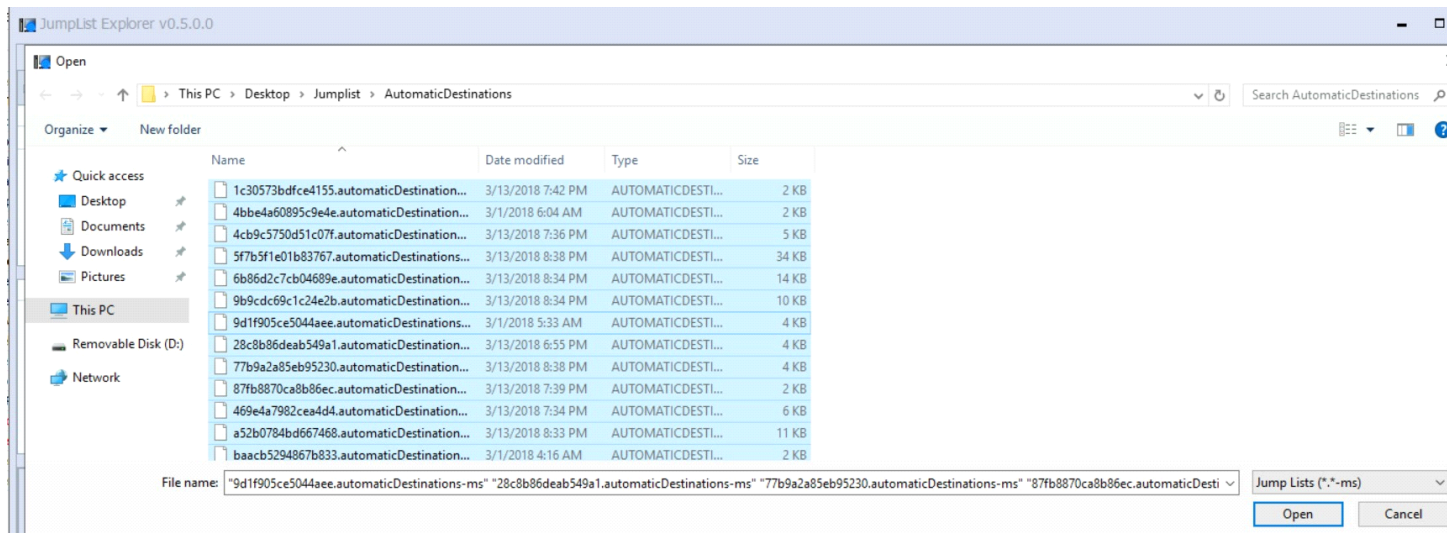
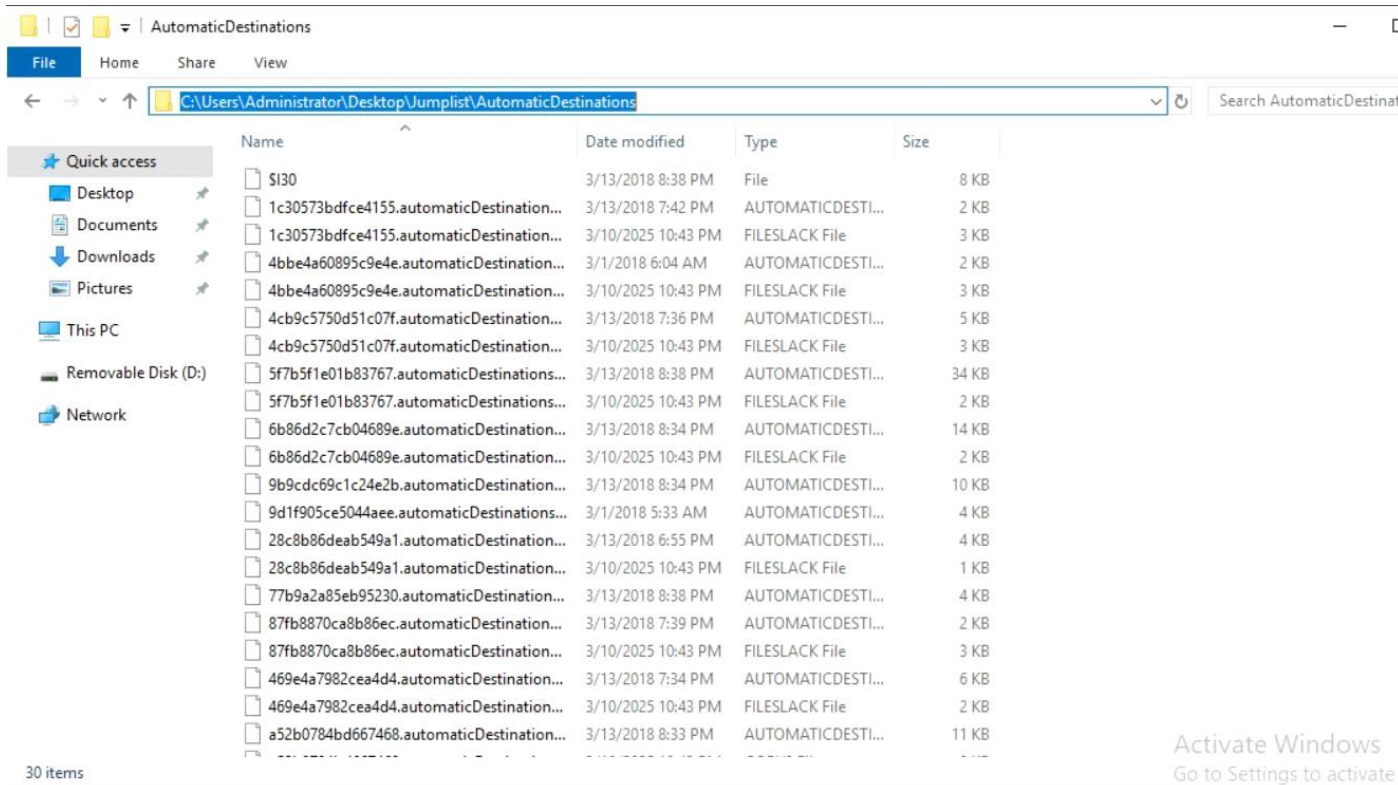
Properties

Name	AutomaticDestinations
File Class	Directory
File Size	232
Physical Size	232
Date Accessed	3/13/2018 9:28:11 PM

Properties Hex Value Interpreter Custom Content Sources

File List

Name	Size	Type	Date Modified
\$I30	8	NTFS Index All...	3/13/2018 8:38:11 PM
1c30573bdfce4155.automaticDestination...	2	Regular File	3/13/2018 7:42:01 PM
1c30573bdfce4155.automaticDestination...	3	File Slack	
28c8b86deab549a1.automaticDestination...	4	Regular File	3/13/2018 6:55:46 PM
28c8b86deab549a1.automaticDestination...	1	File Slack	
469e4a7982cea4d4.automaticDestination...	6	Regular File	3/13/2018 7:34:10 PM
469e4a7982cea4d4.automaticDestination...	2	File Slack	
469e4a7982cea4d4.automaticDestination...	2	Regular File	3/1/2018 6:04:00 AM
4bbe4a60895c9e4e.automaticDestination...	3	File Slack	
4bbe4a60895c9e4e.automaticDestination...	5	Regular File	3/13/2018 7:36:52 PM
4bbe4a60895c9e4e.automaticDestination...	3	File Slack	
4bbe4a60895c9e4e.automaticDestination...	34	Regular File	3/13/2018 8:38:11 PM
4bbe4a60895c9e4e.automaticDestination...	2	File Slack	
4bbe4a60895c9e4e.automaticDestination...	14	Regular File	3/13/2018 8:34:34 PM
4bbe4a60895c9e4e.automaticDestination...	2	File Slack	
4bbe4a60895c9e4e.automaticDestination...	4	Regular File	3/13/2018 8:38:11 PM
4bbe4a60895c9e4e.automaticDestination...	2	Regular File	3/13/2018 7:39:40 PM
4bbe4a60895c9e4e.automaticDestination...	3	File Slack	
4bbe4a60895c9e4e.automaticDestination...	10	Regular File	3/13/2018 8:34:12 PM
4bbe4a60895c9e4e.automaticDestination...	4	Regular File	3/1/2018 5:33:19 AM
4bbe4a60895c9e4e.automaticDestination...	11	Regular File	3/13/2018 8:33:57 PM
4bbe4a60895c9e4e.automaticDestination...		\$I30 INDX Entry	
4bbe4a60895c9e4e.automaticDestination...	2	File Slack	



JumpList Explorer v0.5.0.0

File Help

Drag a column header here to group by that column

Source File Name	Jump List Type	App ID	App ID Description	Lnk File Count	File Size
C:\Users\Administrator\Desktop\JumpList\...	Automatic	1c30573bdfce4155	Unknown Applid	0	0
C:\Users\Administrator\Desktop\JumpList\...	Automatic	4bbe4e60895c9e4e	Unknown Applid	0	0
C:\Users\Administrator\Desktop\JumpList\...	Automatic	4cb9c5750d51c07f	Unknown Applid	2	2
C:\Users\Administrator\Desktop\JumpList\...	Automatic	5f7b5f1e01b83767	Unknown Applid	21	21
C:\Users\Administrator\Desktop\JumpList\...	Automatic	6b86d2c7cb04689e	Unknown Applid	9	9

Name

- 5f7b5f1e01b83767.automaticDestinations-ms
 - Entry #: 0030 - My Computer\C:\Users\dfp\Documents\netscan.xml
 - Entry #: 0029 - My Computer\Z:\Personal\Confidential Document.pdf
 - Entry #: 0028 - My Computer\Z:\Personal\ClassifiedDoc1.pdf
 - Entry #: 0013 - \\SHARES.DFP.EXAM\PUBLIC\Exams\ExamQuestions.pdf
 - Entry #: 0027 - My Computer\Z:\Documents\tools.txt
 - Entry #: 0026 - My Computer\Z:\Documents\Accounts.txt
 - Entry #: 0025 - My Computer\Z:\Tutorials\Lab6\Lab6_2n.ppt

Properties

Appid	5f7b5f1e01b83767
Appid description	Unknown Applid
Pinned count	0
Entries count	21
Last used entry #	30
Version	4

Drag a column header here to group by that column

Entry Num...	Target Created On	Target Modified On	Target Accessed On	Absolute Path
30	2018-03-01 06:03:14	2018-03-01 06:03:14	2018-03-01 06:03:14	My Computer\C:\Users\dfp\Documents\netscan.xml
29	2016-06-21 01:59:27	2016-06-21 01:59:27	2018-03-13 16:34:32	My Computer\Z:\Personal\Confidential Document.pdf
28	2018-03-11 15:23:20	2018-03-11 15:23:20	2018-03-13 16:34:27	My Computer\Z:\Personal\ClassifiedDoc1.pdf
13	2018-03-11 15:55:54	2018-03-11 15:55:54	2018-03-13 15:33:04	\\SHARES.DFP.EXAM\PUBLIC\Exams\ExamQuestions.pdf
27	2016-06-21 09:21:53	2016-06-21 09:21:53	2018-03-13 15:35:08	My Computer\Z:\Documents\tools.txt
26	2016-06-21 01:46:16	2016-06-21 01:46:16	2018-03-13 15:35:08	My Computer\Z:\Documents\Accounts.txt
25	2018-02-07 00:36:11	2018-02-07 00:36:11	2018-03-13 15:35:44	My Computer\Z:\Tutorials\Lab6\Lab6_2g.png
24	2018-03-13 19:42:01	2018-03-13 19:42:04	2018-03-13 19:42:02	My Computer\Documents\Nmap\ScanReport.pdf
23	2018-03-01 06:08:03	2018-03-01 06:08:03	2018-03-01 06:08:03	My Computer\C:\Users\dfp\Downloads\Info_7z
22	2018-03-13 19:36:32	2018-03-11 16:48:06	2018-03-13 19:36:32	My Computer\C:\Users\dfp\Videos\Copied\lamp
21	2018-03-13 19:36:32	2018-03-11 16:43:11	2018-03-13 19:36:32	My Computer\C:\Users\dfp\Videos\Copied\8200
20	2018-03-13 19:35:46	2018-02-07 00:04:28	2018-03-13 19:35:46	My Computer\C:\Users\dfp\Tutorials\Lab6\Lab6_
19	2018-03-13 19:35:11	2016-06-21 09:21:53	2018-03-13 19:35:11	My Computer\C:\Users\dfp\Documents\FromPub

Questions summary

Q: 20/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. What was the most launched application?

Lab questions

Pinned

Answered

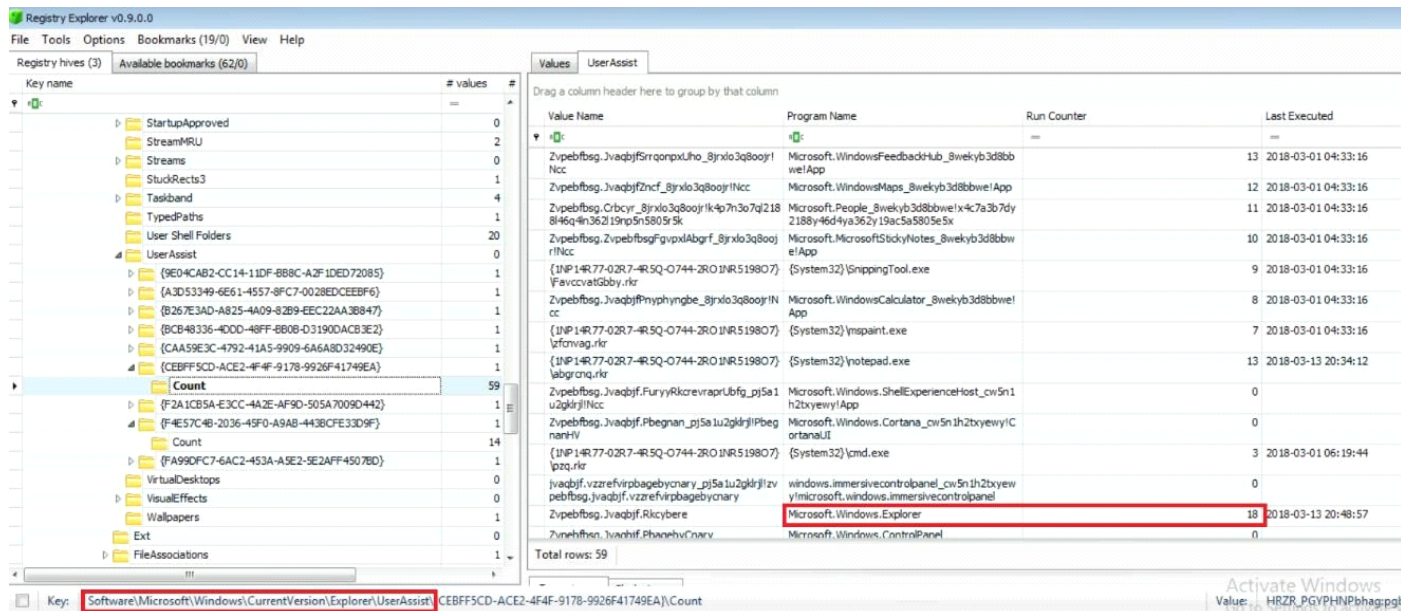
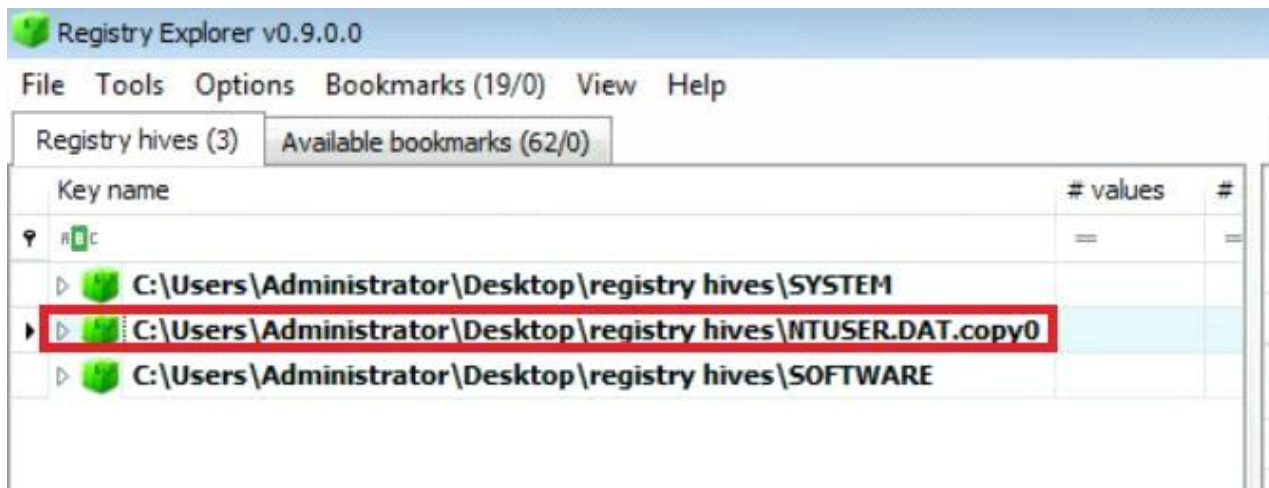
Unanswered

☐ Nmap

☒ Microsoft.Windows.Explorer

☐ notepad.exe

☐ evince



Note: I am not sure from the answer of below questions (21).

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 21/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk2.001. What was the disk schema?

- ☐ GPT
- ☐ SHA256
- ☐ No disk schema
- ☒ MBR

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 22/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk2.001. What part of MBR was damaged?

- ☐ Only the second partition entry for incriminating data was wiped
- ☐ The whole 512 bytes was wiped
- ☐ Only the Partitions section was wiped
- ☒ Only the Bootcode section was wiped

The screenshot shows a Windows XP desktop with a taskbar containing icons for Internet Explorer, HxD, and a folder named 'Exam_Files\Disks\Disk1.001'. The HxD application is open, displaying the contents of 'Disk1.001'. The main window shows a hex view with columns for Offset (h), hex data, and Decoded text. A context menu is open over the selected area, showing options like Undo, Cut, Copy, Paste insert, Paste write, Delete, Fill selection..., Select block..., Select all, and Copy offset. The data inspector on the right shows various data types and their values, including binary, integers, and floating-point numbers.

Hex Editor Data:

Offset (h)	Hex Data	Decoded text
3FFFFDF0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFE00	45 46 49 20 50 41 52 54 00 00 00 01 00 5C 00 00 00	EFI PART....\...
3FFFFE10	5E 7E F1 04 00 00 00 00 00 FF FF 1F 00 00 00 00 00	~ã....yy.....
3FFFFE20	01 00 00 00 00 00 00 00 00 22 00 00 00 00 00 00	"
3FFFFE30	DE FF 1F 00 00 00 00 00 00 FE E0 09 CD EC 83 A7 45	ÿy.....pà.îifSE
3FFFFE40	93 6C 8E 88 C2 9B D6 32 DF FF 1F 00 00 00 00 00	*1ž·Â>ô2By.....
3FFFFE50	80 00 00 00 80 00 00 00 00 28 C2 C1 AD 00 00 00	E...ë... (ÄÄ.....
3FFFFE60	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFE70	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFE80	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFE90	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFEA0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFEB0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFEC0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFED0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFEE0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFEF0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF00	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF10	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF20	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF30	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF40	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF50	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF60	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF70	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF80	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFF90	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFFA0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFFB0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFFC0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
3FFFFFD0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Data Inspector:

Binary (8 bit)	Value
Int8	01000101
UInt8	69
Int16	69
UInt16	17989
Int24	17989
UInt24	4802117
Int32	4802117
UInt32	541673029
Int64	541673029
UInt64	6075990659671082565
LEB128	-59
ULEB128	69
AnsiChar / char8_t	E
WideChar / char16_t	櫻
UTF-8 code point	E (U+0045)
Single (float32)	1.70486124773727E-19
Double (float64)	1.55970874485914E98
OLETIME	Invalid
FILETIME	Invalid

Byte order: ☒ Little endian ☐ Big endian

☐ Hexadecimal basis (for integral numbers)

Then paste them after 512 (After **55 AA Hex**) in the beginning of the Disk

HxD - [C:\DFP\Exam_Files\Disks\Disk1.001]

File Edit Search View Analysis Tools Window Help

16 Windows (ANSI) hex

Disk1.001

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000150	10	EB	F2	F4	EB	FD	2B	C9	E4	64	EB	00	24	02	E0	F8	.ëòôëý+Éädē.\$.àø
00000160	24	02	C3	49	6E	76	61	6C	69	64	20	70	61	72	74	69	\$.ÄInvalid parti
00000170	74	69	6F	6E	20	74	61	62	6C	65	00	45	72	72	6F	72	tion table.Error
00000180	20	6C	6F	61	64	69	6E	67	20	6F	70	65	72	61	74	69	loading operati
00000190	6E	67	20	73	79	73	74	65	6D	00	4D	69	73	73	69	6E	ng system.Missin
000001A0	67	20	6F	70	65	72	61	74	69	6E	67	20	73	79	73	74	g operating syst
000001B0	65	6D	00	00	00	63	7B	9A	00	00	00	00	00	00	00	00	em...c{š.....
000001C0	02	00	EE	FF	FF	FF	01	00	00	00	FF	FF	FF	FF	00	00	..îÿÿÿ....ÿÿÿÿ..
000001D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000001F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55	AA	U²
00000200	45	46	49	20	50	41	52	54	00	00	01	00	5C	00	00	00	EFI PART....\...
00000210	5E	7E	F1	04	00	00	00	00	FF	FF	1F	00	00	00	00	00	^~ñ.....ÿÿ.....
00000220	01	00	00	00	00	00	00	00	22	00	00	00	00	00	00	00	".....
00000230	DE	FF	1F	00	00	00	00	00	FE	E0	09	CD	EC	83	A7	45	pÿ.....pà.îifSE
00000240	93	6C	8E	88	C2	9B	D6	32	DF	FF	1F	00	00	00	00	00	"lŽ^Â>Ö2Bÿ.....
00000250	80	00	00	00	80	00	00	00	28	C2	C1	AD	00	00	00	00	€...€... (ÂÂ.....
00000260	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000270	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000280	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000290	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000002F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000300	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000310	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000320	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000330	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

Then save as the disk (ex. Disk1-fixed.001)

After we fixed the Disk 1

WinHex - [Disk1-Fixed.001]

File Edit Search Navigation View Tools Specialist Options Window Help

Case Data

File Edit

Disk1-Fixed.001

Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	ANSI ASCII	
0000000240	43	50	41	75	32	81	F9	02	01	72	2C	66	68	07	BB	00	CPAu2 ù r,fh »	
0000000256	00	66	68	00	02	00	00	66	68	08	00	00	00	66	53	66	fh fh fsf	
0000000272	53	66	55	66	68	00	00	00	00	66	68	00	7C	00	00	66	SfUfh fh f	
0000000288	61	68	00	00	07	CD	1A	5A	32	F6	EA	00	7C	00	00	CD	ah í Z2ðê í	
0000000304	18	A0	B7	07	EB	08	A0	B6	07	EB	03	A0	B5	07	32	E4	· è ¶ é µ 2ä	
0000000320	05	00	07	8B	F0	AC	3C	00	74	09	BB	07	00	B4	0E	CD	<ð-< t » ' í	
0000000336	10	EB	F2	F4	EB	FD	2B	C9	E4	64	EB	00	24	02	E0	F8	èòðëý+Èädè \$ àø	
0000000352	24	02	C3	49	6E	76	61	6C	69	64	20	70	61	72	74	69	\$ ÅInvalid parti	
0000000368	74	69	6F	6E	20	74	61	62	6C	65	00	45	72	72	6F	72	tion table Error	
0000000384	00	6C	6F	61	64	69	6E	67	20	6F	70	65	72	61	74	69	loading operati	
0000000400	6E	67	20	73	79	73	74	65	6D	00	4D	69	73	73	69	6E	ng system Missin	
0000000416	67	20	6F	70	65	72	61	74	69	6E	67	20	73	79	73	74	g operating syst	
0000000432	65	6D	00	00	00	63	7B	9A	00	00	00	00	00	00	00	00	em c{š	
0000000448	02	00	EE	FF	FF	FF	01	00	00	00	FF	FF	FF	FF	FF	00	00	ïÿÿÿ ÿÿÿÿ
0000000464	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000480	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000496	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55 AA	U*
0000000512	45	46	49	20	50	41	52	54	00	00	01	00	5C	00	00	00	00	EFI PART
0000000528	5E	7E	F1	04	00	00	00	00	FF	FF	1F	00	00	00	00	00	00	^~ñ ÿÿ
0000000544	01	00	00	00	00	00	00	00	22	00	00	00	00	00	00	00	00	"
0000000560	DE	FF	1F	00	00	00	00	00	FE	E0	09	CD	EC	83	A7	45	00	Ëÿ ðà ílfSE
0000000576	93	6C	8E	88	C2	9B	D6	32	DF	FF	1F	00	00	00	00	00	00	"lž~Å>Ö2Bÿ
0000000592	80	00	00	00	80	00	00	00	28	C2	C1	AD	00	00	00	00	00	€ € (ÅÅ-
0000000608	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000624	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000640	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

Page 1 of 2,581,112 Offset: 384 = 32 | Block: n/a | Size: av

WinHex - [Disk1-Fixed.001]

File Edit Search Navigation View Tools Specialist Options Window Help

Case Data

File Edit

Disk1-Fixed.001

Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	ANSI ASCII	
0000000240	43	50	41	75	32	81	F9	02	01	72	2C	66	68	07	BB	00	CPAu2 ù r,fh »	
0000000256	00	66	68	00	02	00	00	66	68	08	00	00	00	66	53	66	fh fh fsf	
0000000272	53	66	55	66	68	00	00	00	00	66	68	00	7C	00	00	66	SfUfh fh f	
0000000288	61	68	00	00	07	CD	1A	5A	32	F6	EA	00	7C	00	00	CD	ah í Z2ðê í	
0000000304	18	A0	B7	07	EB	08	A0	B6	07	EB	03	A0	B5	07	32	E4	· è ¶ é µ 2ä	
0000000320	05	00	07	8B	F0	AC	3C	00	74	09	BB	07	00	B4	0E	CD	<ð-< t » ' í	
0000000336	10	EB	F2	F4	EB	FD	2B	C9	E4	64	EB	00	24	02	E0	F8	èòðëý+Èädè \$ àø	
0000000352	24	02	C3	49	6E	76	61	6C	69	64	20	70	61	72	74	69	\$ ÅInvalid parti	
0000000368	74	69	6F	6E	20	74	61	62	6C	65	00	45	72	72	6F	72	tion table Error	
0000000384	00	6C	6F	61	64	69	6E	67	20	6F	70	65	72	61	74	69	loading operati	
0000000400	6E	67	20	73	79	73	74	65	6D	00	4D	69	73	73	69	6E	ng system Missin	
0000000416	67	20	6F	70	65	72	61	74	69	6E	67	20	73	79	73	74	g operating syst	
0000000432	65	6D	00	00	00	63	7B	9A	00	00	00	00	00	00	00	00	em c{š	
0000000448	02	00	EE	FF	FF	FF	01	00	00	00	FF	FF	FF	FF	FF	00	00	ïÿÿÿ ÿÿÿÿ
0000000464	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000480	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000000496	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	55 AA	U*

Page 1 of 2,581,112 Offset: 384 = 32 | Block: n/a | Size: av

Specialist

- Refine Volume Snapshot... F10
- Technical Details Report... Ctrl+F10
- Interpret Image File As Disk
- Mount as Drive Letter... ||+F10
- Reconstruct RAID System...
- Gather Free Space...
- Gather Slack Space...
- Gather Inter-Partition Space...
- Gather Text...
- Bates Number Files...
- Trusted Download...

WinHex - [Disk1-Fixed.001]

File Edit Search Navigation View Tools Specialist Options Window Help

Case Data

File Edit

Disk1-Fixed

Partitioning style: GPT

Name	Ext.	Size	Created	Modified	Record changed	Attr.	1st sector
Start sectors		17.0 KB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Partition 1	?	32.0 MB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Partition 4	NTFS	450 MB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Partition 3	NTFS	50.0 MB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Partition 2	?	450 MB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Partition gap		47.0 KB	out of bounds ↑	out of bounds ↑	out of bounds ↑		
Unpartitioned space		41.9 MB	out of bounds ↑	out of bounds ↑	out of bounds ↑		

Offset: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

0000000000 83 C0 8E D0 BC 00 7C 8E C0 8E D8 BE 00 7C BF 00 3A ŽĐ 4 | ŽA Ž0 4 | ž

0000000016 06 B9 00 02 FC F3 A4 50 68 1C 06 CB FB B9 04 00 2 uóMPh ÈÙ²

0000000032 BD BE 07 80 7E 00 00 7C 0B 0F 85 0E 01 83 C5 10 ¼ e~ | ... fĀ

0000000048 E2 F1 CD 18 88 56 00 55 C6 46 11 05 C6 46 10 00 aňĭ ^V UEF EF

0000000064 B4 41 BB AA 55 CD 13 5D 72 0F 81 FB 55 AA 75 09 'A»*UÍ]r ŪU²u

0000000080 F7 C1 01 00 74 03 FE 46 10 66 60 80 7E 10 00 74 ÷Ā t pF f' e~ t

0000000096 26 66 68 00 00 00 00 66 FF 76 08 68 00 00 68 00 &fh fÿv h h

0000000112 7C 68 01 00 68 10 00 B4 42 8A 56 00 8B F4 CD 13 |h h 'BŠV <óĭ

0000000128 9F 83 C4 10 9E EB 14 B8 01 02 BB 00 7C 8A 56 00 ŸfĀ žē . » |ŠV

0000000144 8A 76 01 8A 4E 02 8A 6E 03 CD 13 66 61 73 1C FE Šv ŠN Šn ĭ fas p

0000000160 4E 11 75 0C 80 7E 00 80 0F 84 8A 00 B2 80 EB 84 N u e~ e „Š 'eē„

0000000176 55 32 E4 8A 56 00 CD 13 5D EB 9E 81 3E FE 7D 55 U2āšV ĭ jēž >p}U

0000000192 AA 75 6E FF 76 00 E8 8D 00 75 17 FA B0 D1 E6 64 *unÿv è u ū°Næd

0000000208 E8 83 00 B0 DF E6 60 E8 7C 00 B0 FF E6 64 E8 75 èf °Bæ'è | °ÿædèu

Sector 0 of 2,097,153 | Offset: 0 | = 51 | Block: n/a | Size:

WinHex - [Disk1-Fixed.001]

File Edit Search Navigation View Tools Specialist Options Window Help

Case Data

File Edit

Disk1-Fixed

Partitioning style: GPT

View

- ☐ Text Display Only F7
- ☐ Hex Display Only Alt+F7
- Character set No. 1 ð+F7 >
- Character set No. 2 >
- Character set No. 3 >
- Record Presentation... >
- Show >
- Template Manager... Alt+F12
- Tables >
- One Column Less
- One Column More
- ☐ Synchronize Windows
- ☐ Synchronize & Compare
- ☒ Refresh View

Created	Modified	Record changed	Attr.	1st sector
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		
out of bounds ↑	out of bounds ↑	out of bounds ↑		

Offset: 11 12 13 14 15

BE 00 7C BF 00 3A ŽĐ 4 | ŽA Ž0 4 | ž

CB FB B9 04 00 2 uóMPh ÈÙ²

0E 01 83 C5 10 ¼ e~ | ... fĀ

05 C6 46 10 00 aňĭ ^V UEF EF

FB 55 AA 75 09 'A»*UÍ]r ŪU²u

F7 C1 01 00 74 03 FE 46 10 66 60 80 7E 10 00 74 ÷Ā t pF f' e~ t

26 66 68 00 00 00 00 66 FF 76 08 68 00 00 68 00 &fh fÿv h h

Sector 0 of 2,097,153 | Offset: 0 | = 51 | Block: n/a | Size:

Partitioning style

Name

Start sectors
Partition 1
Partition 4
Partition 3
Partition 2
Partition gap
Unpartitioned

Offset

0000000000
0000000016
0000000032
0000000048
0000000064
0000000080
0000000096
0000000112
0000000128
0000000144
0000000160
0000000176
0000000192
0000000208

Title	Description	Name	Time
Boot Sector FAT	BIOS parameter block (BPB) and more	Boot Sector F...	02/05/201...
Boot Sector FAT32	BIOS parameter block (BPB) and more	Boot Sector F...	02/05/201...
Boot Sector NTFS	Boot sector of an NTFS partition	Boot Sector N...	02/05/201...
Ext2/Ext3 Directory Entry	Locates the Inode for a given filename	Ext Directory ...	02/05/201...
Ext2/Ext3 Group Descriptor	Locates the meta blocks for a block group	Ext Group De...	02/05/201...
Ext2/Ext3 Inode	Contains a file's meta information (classic block formatting)	Ext Inode.tpl	02/05/201...
Ext2/Ext3/Ext4 Superblock	To be applied to offset 1024 of an Ext2/3/4 partition	Ext Superbloc...	02/05/201...
FAT Directory Entry	Normal/short entry format	FAT Directory ...	02/05/201...
FAT Directory Entry	Long entry format	FAT L EN Entr	02/05/201...
GUID Partition Table	Contains GUID partition table	GUID Partition...	02/05/201...
HFS+ Volume Header	Located 1024 bytes from the start of the volume	HFS+ Volume ...	02/05/201...
Master Boot Record	Contains partition table	Master Boot R...	02/05/201...
NTFS FILE Record	To be applied to records in the Master File Table	NTFS FILE R...	02/05/201...



Apply!

Close



New...



Edit...



Delete...

AA 75 6E FF 76 00 E8 8D 00 75 17 FA B0 D1 E6 64
E8 83 00 B0 DF E6 60 E8 7C 00 B0 FF E6 64 E8 75

AA 75 6E FF 76 00 E8 8D 00 75 17 FA B0 D1 E6 64
E8 83 00 B0 DF E6 60 E8 7C 00 B0 FF E6 64 E8 75

*unÿv è u ú°Næd
èf °Bæ°è| °ÿæðeu

0+0+3 file

1st sector

[C:\DFP\Exam_Files\]

Default Edit Mode
State:Undo level:
Undo reverses:Total capacity:
1,073,7

Bytes per sector:

Partition:
Relative sector No.:Mode:
Offsets:

HEX GUID Partition Table, Base Offset: 0

452	Ending Sector	FF
453	Ending Cylinder	FF
454	Starting LBA	1
458	Size in LBA	4,294,967,295

GUID Partition Table Header

512	Signature (must be 45[.. ₅₄)	45 46 49 20 50 41 52 54
520	Revision No	00 00 01 00
524	Header Size (should be 92)	92
528	Header CRC32	4F17E5E
536	Primary LBA (should be 1)	2,097,151
544	Backup LBA	1
552	First Usable LBA	34
560	Last Usable LBA	2,097,118
568	Disk GUID	FE E0 09 CD EC 83 A7 45 93 6C 8E 88 C2 9B D6 32
568	Disk GUID	{CD09E0FE-83EC-45A7-936C-8E88C29BD632}
584	Partition Entry LBA (should be 2)	2,097,119
592	MaxNo of Partition Entries	128
596	Size of Partition Entry	128
600	Partition Entry Array CRC32	ADC1C228

GUID Partition Table Entry #1

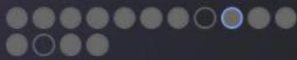
78 00 00 00 66 53 66 | fh fh fsh

Activate Windows

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 24/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk1.001. What was the label of the drive holding the incriminating data (kittens represent illegal material)?



NTFS



MyFiles



Data



Tools

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 25/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk1.001. Is there any illegal material (kittens represent illegal material)?

- ☐ Malicious EXE files
- ☐ Illegal book copies
- ☒ Illegal material (kittens)
- ☐ Stolen documents

Questions summary



Lab questions

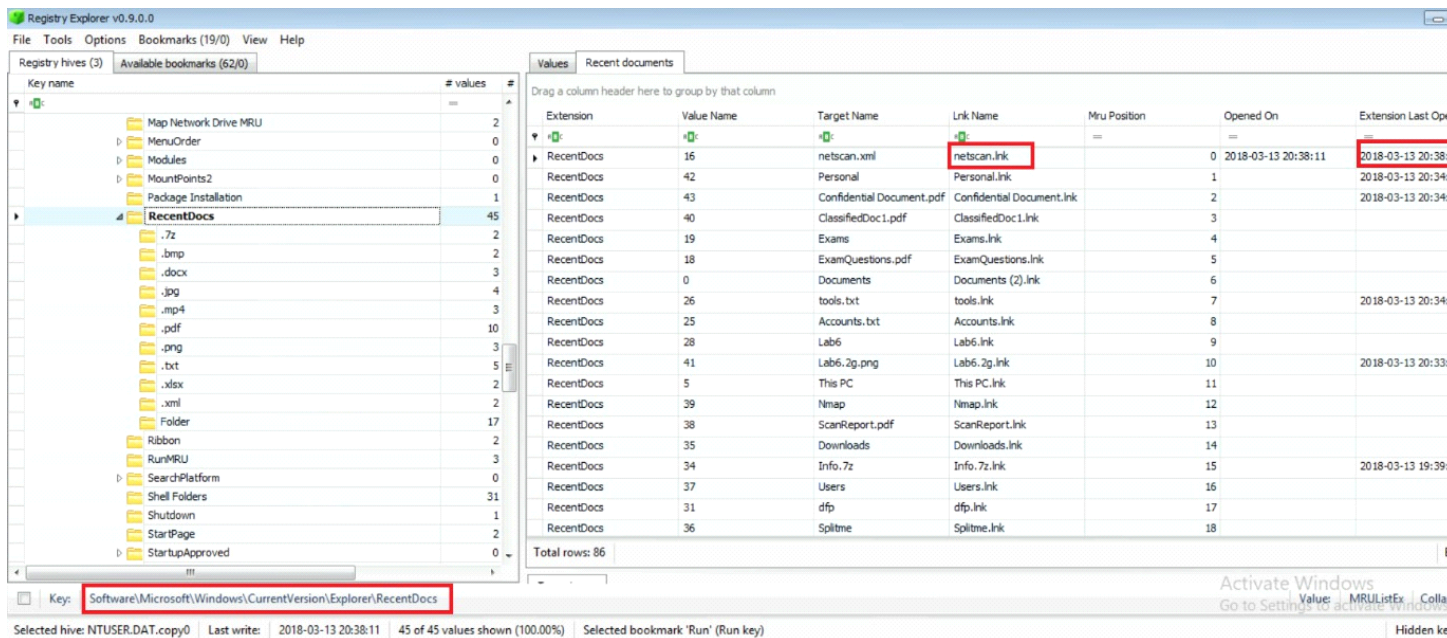
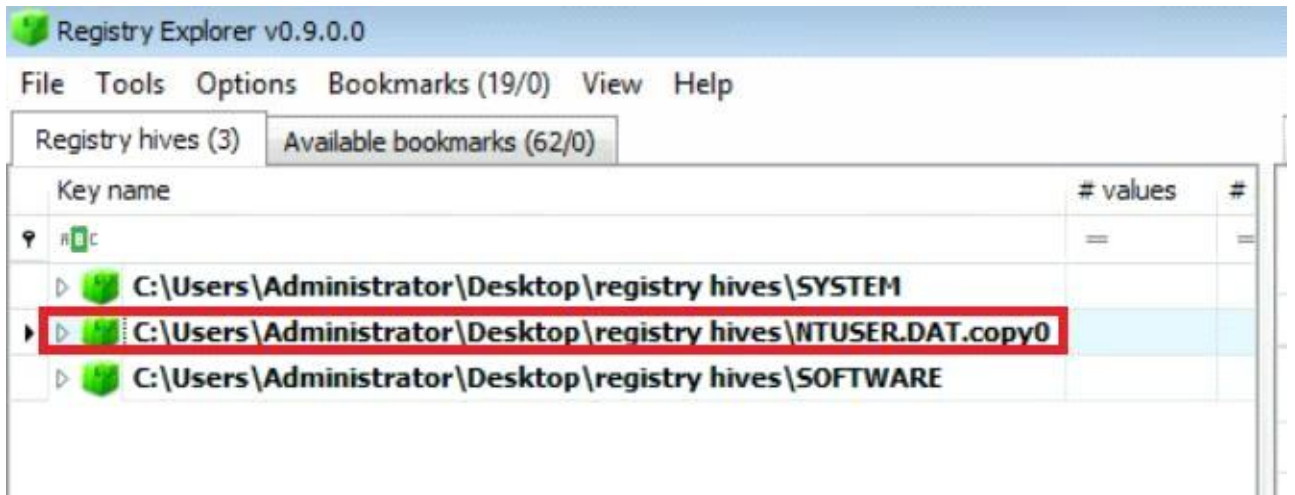


- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 26/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. When was the file netscan.xml last opened?

- ☐ 2018-03-13 20:34:34
- ☐ 2018-03-13 20:34:12
- ☒ 2018-03-13 20:38:11
- ☐ 2018-03-01 5:34:51



Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 27/30

Log into the 172.16.90.101 machine. Analyze the B1 and B2 PCAP files located at /home/elsuser/Desktop/Exam_Files/Pcaps/. What is the name of the infected host?

☐ Christyn-Lyi-PC

☒ Chris-Lyons-PC

☐ Charla-swan-PC

☐ Charlos-lyri-PC

B1.pcap					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000	10.1.1.97	224.0.0.252	LLMNR	74 Standard query 0xde83 ANY Chris-Lyons-PC
2	0.057148	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<00>
3	0.057280	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<20>
4	0.057280	10.1.1.97	10.1.1.255	NBNS	110 Registration NB WORKGROUP<00>
5	0.103416	10.1.1.97	224.0.0.252	LLMNR	74 Standard query 0xde83 ANY Chris-Lyons-PC
6	0.253799	10.1.1.97	10.1.1.1	DNS	91 Standard query 0x2933 SRV _ldap._tcp.dc._msdcs.mshome.net
7	0.254047	10.1.1.1	10.1.1.97	DNS	91 Standard query response 0x2933 No such name SRV _ldap._tcp.dc._msdcs.mshome.net
8	0.255556	10.1.1.97	10.1.1.1	DNS	91 Standard query 0x519e SRV _ldap._tcp.dc._msdcs.mshome.net
9	0.255774	10.1.1.1	10.1.1.97	DNS	91 Standard query response 0x519e No such name SRV _ldap._tcp.dc._msdcs.mshome.net
10	0.821017	10.1.1.97	10.1.1.255	NBNS	110 Registration NB WORKGROUP<00>
11	0.821017	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<20>
12	0.821018	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<00>
13	1.585428	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<00>
14	1.585429	10.1.1.97	10.1.1.255	NBNS	110 Registration NB CHRIS-LYONS-PC<20>
Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0 Ethernet II, Src: ASUSTekC_d4:9a:e7 (00:22:15:d4:9a:e7), Dst: IPv4mcast_fc (01:00:5e:00:00:fc) Internet Protocol Version 4, Src: 10.1.1.97, Dst: 224.0.0.252 User Datagram Protocol, Src Port: 61978, Dst Port: 5355					

○ Unanswered

Log into the 172.16.90.101 machine. Analyze the A1 up to A6 PCAP files located at /home/elsuser/Desktop/Exam_Files/Pcaps/. What is the IP address of the compromised site?

62.71.34.33

Wireshark packet capture showing DNS traffic. The packet list shows a query for fonts.gstatic.com and a response. The packet details pane shows the query and response structure. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
249	51.741509	172.16.103.77	172.16.103.1	DNS	77	Standard query 0x7bf6 A fonts.gstatic.com
250	51.744759	172.16.103.77	172.16.103.1	DNS	75	Standard query 0xe955 A ssl.gstatic.com
255	51.813230	172.16.103.1	172.16.103.77	DNS	129	Standard query response 0x7bf6 A fonts.gstatic.com CNAME gstaticadssl.l.google.com
259	51.853255	172.16.103.1	172.16.103.77	DNS	91	Standard query response 0xe955 A ssl.gstatic.com A 172.217.0.227
595	52.495727	172.16.103.77	172.16.103.1	DNS	75	Standard query 0xca0d A www.gstatic.com
629	52.584117	172.16.103.1	172.16.103.77	DNS	91	Standard query response 0xca0d A www.gstatic.com A 172.217.0.227
753	52.822357	172.16.103.77	172.16.103.1	DNS	84	Standard query 0x01de A www.googleadservices.com
783	52.894702	172.16.103.77	172.16.103.1	DNS	75	Standard query 0x43b3 A apis.google.com
788	52.919565	172.16.103.1	172.16.103.77	DNS	138	Standard query response 0x01de A www.googleadservices.com CNAME pagead.l.doubleclick.net
792	52.975244	172.16.103.1	172.16.103.77	DNS	112	Standard query response 0x43b3 A apis.google.com CNAME plus.l.google.com A 172.217.0.227
885	53.673186	172.16.103.77	172.16.103.1	DNS	80	Standard query 0x284d A adservice.google.com
888	53.818454	172.16.103.1	172.16.103.77	DNS	136	Standard query response 0x284d A adservice.google.com CNAME pagead46.l.doubleclick.net
933	58.386997	172.16.103.77	172.16.103.1	DNS	73	Standard query 0x4765 A lgrrealty.com
942	58.618723	172.16.103.1	172.16.103.77	DNS	89	Standard query response 0x4765 A lgrrealty.com A 64.71.35.26
952	59.081909	172.16.103.77	172.16.103.1	DNS	77	Standard query 0x8343 A www.lgrrealty.com
955	59.262868	172.16.103.1	172.16.103.77	DNS	93	Standard query response 0x8343 A www.lgrrealty.com A 64.71.35.26
979	59.624927	172.16.103.77	172.16.103.1	DNS	87	Standard query 0x6fd3 A sooncalling120112345678.tk
1124	59.921997	172.16.103.1	172.16.103.77	DNS	103	Standard query response 0x6fd3 A sooncalling120112345678.tk A 162.244.35.33
1158	60.190671	172.16.103.77	172.16.103.1	DNS	80	Standard query 0x8707 A istupport6191112.tk

Frame 942: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface 0
 Ethernet II, Src: Avaya_54:F9:28 (c8:f4:06:54:f9:28), Dst: Msi_d8:69:3a (00:16:17:d8:69:3a)
 Internet Protocol Version 4, Src: 172.16.103.1, Dst: 172.16.103.77
 User Datagram Protocol, Src Port: 53, Dst Port: 61738
 Domain Name System (response)

A3.pcap						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
http and ip.addr == 64.71.35.26						
No.	Time	Source	Destination	Protocol	Length	Info
947	58.753295	172.16.103.77	64.71.35.26	HTTP	456	GET / HTTP/1.1
953	59.114829	64.71.35.26	172.16.103.77	HTTP	60	HTTP/1.1 302 Moved Temporarily (text/html)
959	59.310996	172.16.103.77	64.71.35.26	HTTP	460	GET / HTTP/1.1
987	59.681136	64.71.35.26	172.16.103.77	HTTP	1140	HTTP/1.1 200 OK (text/html)
989	59.682741	172.16.103.77	64.71.35.26	HTTP	419	GET /wp-content/themes/workplace/style.css HTTP/1.1
1003	59.721356	172.16.103.77	64.71.35.26	HTTP	416	GET /wp-content/themes/workplace/scripts/suckerfish.js HTTP/1.1
1005	59.721620	172.16.103.77	64.71.35.26	HTTP	416	GET /wp-content/themes/workplace/scripts/jquery.min.js HTTP/1.1
1007	59.721973	172.16.103.77	64.71.35.26	HTTP	418	GET /wp-content/themes/workplace/scripts/jquery.cycle.js HTTP/1.1
1009	59.722119	172.16.103.77	64.71.35.26	HTTP	422	GET /wp-content/themes/workplace/scripts/jquery.functions.js HTTP/1.1
1011	59.722415	172.16.103.77	64.71.35.26	HTTP	423	GET /wp-content/themes/workplace/scripts/jquery.dimensions.js HTTP/1.1
1033	59.797871	64.71.35.26	172.16.103.77	HTTP	856	HTTP/1.1 200 OK (text/css)
1035	59.799697	172.16.103.77	64.71.35.26	HTTP	417	GET /wp-content/themes/workplace/scripts/jquery.fade.js HTTP/1.1
1037	59.819964	64.71.35.26	172.16.103.77	HTTP	1013	HTTP/1.1 200 OK (application/javascript)
1048	59.823291	172.16.103.77	64.71.35.26	HTTP	420	GET /wp-content/themes/workplace/scripts/jquery.tooltip.js HTTP/1.1
1053	59.848227	64.71.35.26	172.16.103.77	HTTP	583	HTTP/1.1 200 OK (application/javascript)
1099	59.899796	64.71.35.26	172.16.103.77	HTTP	473	HTTP/1.1 200 OK (application/javascript)
1100	59.900075	64.71.35.26	172.16.103.77	HTTP	1432	HTTP/1.1 200 OK (application/javascript)
1114	59.913080	64.71.35.26	172.16.103.77	HTTP	175	HTTP/1.1 200 OK (application/javascript)
1121	59.913868	64.71.35.26	172.16.103.77	HTTP	268	HTTP/1.1 200 OK (application/javascript)
1146	59.958981	64.71.35.26	172.16.103.77	HTTP	829	HTTP/1.1 200 OK (application/javascript)

ine.attackdefensecloudlabs.com/overlay/64cb52d133c2ee146f7df77957a50b9940131c756dfdd849b4c287307c897859

Windows10 Linux

A3.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request.full.uri contains "lgrrealty.com"

No.	Time	Source	Destination	Protocol	Length	Info
947	58.753295	172.16.103.77	64.71.35.26	HTTP	456	GET / HTTP/1.1
959	59.310996	172.16.103.77	64.71.35.26	HTTP	460	GET / HTTP/1.1
989	59.682741	172.16.103.77	64.71.35.26	HTTP	419	GET /wp-con
1003	59.721356	172.16.103.77	64.71.35.26	HTTP	416	GET /wp-con
1005	59.721620	172.16.103.77	64.71.35.26	HTTP	416	GET /wp-con
1007	59.721973	172.16.103.77	64.71.35.26	HTTP	418	GET /wp-con
1009	59.722119	172.16.103.77	64.71.35.26	HTTP	422	GET /wp-con
1011	59.722415	172.16.103.77	64.71.35.26	HTTP	423	GET /wp-con
1035	59.799697	172.16.103.77	64.71.35.26	HTTP	417	GET /wp-con
1048	59.823291	172.16.103.77	64.71.35.26	HTTP	420	GET /wp-con

- Mark/Unmark Packet(s) Ctrl+M
- Ignore/Unignore Packet(s) Ctrl+D
- Set/Unset Time Reference Ctrl+T
- Time Shift... Ctrl+Shift+T
- Packet Comments
- Edit Resolved Name
- Apply as Filter
- Prepare as Filter
- Conversation Filter
- Colorize Conversation
- SCTP
- Follow TCP Stream Ctrl+Alt+Shift+T
- Copy UDP Stream Ctrl+Alt+Shift+U
- Protocol Preferences DCCP Stream Ctrl+Alt+Shift+F
- Decode As... TLS Stream Ctrl+Alt+Shift+S
- Show Packet in New Window HTTP Stream Ctrl+Alt+Shift+H
- HTTP/2 Stream
- QUIC Stream
- SIP Call

Frame 947: 456 bytes on wire (3648 bits), 456 bytes captured (3648 bits)

Ethernet II, Src: Msi_d8:69:3a (08:16:17:d8:69:3a), Dst: Avaya_54:f9:28 (c8:f4:06:54:f9:28)

Internet Protocol Version 4, Src: 172.16.103.77, Dst: 64.71.35.26

Transmission Control Protocol, Src Port: 49169, Dst Port: 80, Seq: 1, Ack: 1, Len: 402

0000 c8 f4 06 54 f9 28 00 16 17 d8 69 3a 08 00 45 00 ...T(. . . i: .E.

C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-5613-1001

Recycle bin path: 'D:\S-1-5-21-319621592-4104958386-1817415613-1001'

Version: 2

Index	Deleted	Time	Size	Path
		2018-03-01 05:49:21	1107	C:\Users\dfp\Desktop\Quick Stego.lnk
\$1B700LU.lnk		2018-03-13 19:42:26	153143	C:\Users\dfp\Documents\netscan.xml
\$15VZ1UG.xml		2018-03-13 19:42:44	13195	C:\Users\dfp\Documents\FromPublic\ExamQuestions.docx
\$1D1X4E3.docx		2018-03-13 19:45:17	3240054	C:\Users\dfp\Desktop\CyberAttack-Info.bmp
\$1U878YA.bmp		2018-03-13 19:45:54	531368	C:\Users\dfp\Desktop\Utils\putty.exe
\$1Q0SPGZ.exe		2018-03-13 19:45:57	1541744	C:\Users\dfp\Desktop\Utils\splitme.exe
\$1012QIU.exe		2018-03-13 19:45:58	461952	C:\Users\dfp\Desktop\Utils\runtime.exe
\$1KS6DLY.exe		2018-03-13 20:34:57	359336	C:\Users\dfp\Desktop\Utils
\$1GD3V7R		2018-03-13 20:34:59	865	C:\Users\Public\Desktop\notepad++.lnk
\$1ORULNX.lnk		2018-03-13 20:37:15	1907	C:\Users\dfp\Documents\Nmap\network-scan.xml
\$1D2B777.xml				

C:\DFP\Tools\System & Network Forensics\rifiuti2-0.6.1-win\x64>rifiuti-vista.exe D:\S-1-5-21-319621592-4104958386-5613-1001_

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 30/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. When was the network share "\shares.dfp.exam\" opened?

☒ 2018-03-01 5:45:28

☐ 2017-03-13 20:34:12

☐ 2018-03-13 20:34:34

☐ 2018-03-13 20:31:34

Registry Explorer v0.9.0.0

File Tools Options Bookmarks (19/0) View Help

Registry hives (3)

Available bookmarks (62/0)

Key name	# values	#
 C:\	=	=
▶  C:\Users\Administrator\Desktop\registry hives\SYSTEM		
▶  C:\Users\Administrator\Desktop\registry hives\NTUSER.DAT.copy0		
▶  C:\Users\Administrator\Desktop\registry hives\SOFTWARE		

Registry Explorer v0.9.0.0

File Tools Options Bookmarks (19/0) View Help

Registry hives (3) Available bookmarks (62/0)

Key name # values #

- CabinetState 2
- CD Burning 0
- CIDOpen 0
- CIDSave 0
- CLSID 0
- ComDlg32 0
- Desktop 0
- Discardable 0
- FileExts 0
- HideDesktopIcons 0
- LogonStats 2
- LowRegistry 0
- Map Network Drive MRU 2
- MenuOrder 0
- Modules 0
- MountPoints2 0
- Package Installation 1
- RecentDocs 45
- Ribbon 2
- RunMRU 3**
- SearchPlatform 0
- Shell Folders 31
- Shutdown 1
- StartPage 2

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU

Selected hive: NTUSER.DAT.copy0 | Last write: 2018-03-01 05:45:28 | 3 of 3 values shown (100.00%) | Selected bookmark 'Run' (Run key)

Values RunMRU

Drag a column header here to group by that column

Value Name	Mru Position	Executable	Opened On
a	0	\\shares.dfp.exam\	2018-03-01 05:45:28
b	1	calc	

Total rows: 2

Activate Windows
Go to Settings to activate Windows

Domains

Overall required score: 76%

Your score: 90%

Passed

Evidence Preservation

Your score: 100%

Understand the methodology of collection	2/2
Preserve and maintain the integrity of evidence	2/2
Execute the steps of collection plan based on best practices	2/2

Storage Device Fundamentals

Your score: 66%

Demonstrate how to analyze logical storage characteristics	2/2
Demonstrate how to analyze physical device characteristics	2/4

Fundamentals of Digital Forensics

Your score: 100%

Identify evidence of execution on Windows operating systems	2/2
Understand the basic structure of a digital forensic report	2/2
Analyze digital forensics artifacts on Windows operating systems	6/6

Digital forensic tools and techniques

Your score: 87%

Demonstrate appropriate usage of log and timeline tools and techniques	3/3
Demonstrate appropriate usage of network analysis tools and techniques	2/3
Demonstrate appropriate usage of digital forensic analysis tools and techniques	2/2

- Logical acquisition is taking a forensic copy of the whole ____.
- Folder (directory)
- System
- Disk
- Partition/Volume

■ **Correct Answer: D) Partition/Volume**

- **Why is a write blocker required when acquiring a disk image?**

- To encrypt the data being copied
- To protect the integrity of data being copied
- To protect the disk from damage
- To protect the partition from damage

■ **Correct Answer: B) To protect the integrity of data being copied**

- **Static or dynamic acquisition: Which is better?**

- None of them, actually network acquisitions must be performed first
- It just depends on the case you are dealing with
- Dynamic, because you can get a copy of what is actually running on the system at that time
- Static, because you can take a full forensic image without getting interrupted by a running process

■ **Correct Answer: B) It just depends on the case you are dealing with**

- **To make sure you have a working backup of your evidence in case something happens, it is good to:**

- Have more than one digital forensics lab
- Create more than one copy of the disk evidence
- Have more than one digital forensics toolkit
- Create copies of forensic images

■ **Correct Answer: D) Create copies of forensic images** **Note:** I am not sure from the answer of below questions (21).

- **What evidence could we find in the HKLM\System\MountedDevices key?**

- Volume GUID
- USB entries
- Volume Serial Number
- Device type

■ **Correct Answer: A) Volume GUID**

• **Jumplists could help when you are trying to investigate:**

- What files have been copied or deleted from the system
- What executable files have been executed and from where
- What System files were last used
- What was the user's activity

■ **Correct Answer: D) What was the user's activity**

• **Assume you have a case where a malicious EXE was launched on a system. Which artifact would give you an idea of what files it used and called during its execution?**

- Prefetch files
- AppCompatCache files
- LNK files
- Registry files

■ **Correct Answer: A) Prefetch files**

• **Which is the most difficult and important part when writing your investigation report?**

- Preparing the figures and tables
- Writing the executive summary
- Writing the technical details
- Writing the objectives of the report

■ **Correct Answer: B) Writing the executive summary**

G. The best investigation reports are those that _____.

- Have the best executive summary, because executives are mostly interested in that section
- Explain everything from A to Z
- Are full of details and use long paragraphs to explain their work
- Are easy to understand by anyone related to the investigation

■ Correct Answer: D) Are easy to understand by anyone related to the investigation

• Solid-State Drives use _____ as the smallest readable and writable unit, with the size of _____ bytes.

- Page, 4096
- Block, 4096
- Block, 512
- Page, 512

■ Correct Answer: A) Page, 4096

• When creating a volume with the NTFS file system, by default it will preserve _____ for the MFT zone, so the MFT does not get fragmented:

- 75% of the volume size
- 50% of the volume size
- 25% of the volume size
- 12.5% of the volume size

■ Correct Answer: D) 12.5% of the volume size

The screenshot shows a quiz interface with a dark theme. On the left, there is a sidebar with a 'Questions summary' section showing a row of 15 circles, with the 14th circle highlighted in blue. Below this is a 'Lab questions' section with a row of 15 circles, where the 14th circle is also highlighted in blue. A legend indicates that an orange circle is 'Pinned', a grey circle is 'Answered', and a white circle is 'Unanswered'. The main area on the right shows the question 'Q: 14/30' and the text 'Which of the following protocols below generates GET/POST requests?'. There are four radio button options: IMAP, HTTP (which is selected with a blue dot), FTP, and DNS.

Questions summary

Q: 14/30

Which of the following protocols below generates GET/POST requests?

☐ IMAP

☒ HTTP

☐ FTP

☐ DNS

Lab questions

● Pinned

● Answered

○ Unanswered

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 16/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. What was the timezone of this system?

☐ UTC-9

☐ UTC+8

☐ UTC-8

☒ UTC-7

Questions summary



Lab questions



- ☒ Pinned
- ☒ Answered
- ☐ Unanswered

Q: 17/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. Based on the jumplists identified, was the "ExamQuestions.docx" copied to the suspect's machine?

☐ Yes, a copy was found in the path C:\Users\dfp\Copied

☒ Yes, a copy was found in the path C:\Users\dfp\Documents\FromPublic\

☐ Yes, a copy was found in the path C:\Users\dfp\Documents\Exams\

☐ No, it was only accessed from the network share

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 20/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. What was the most launched application?

- ☐ Nmap
- ☐ evince
- ☒ Microsoft.Windows.Explorer
- ☐ notepad.exe

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 21/30

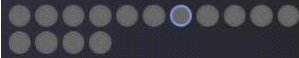
Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk2.001. How many partitions exist on disk?

- ☐ 5
- ☒ 4
- ☐ 3
- ☐ 2

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 22/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk2.001. What is the label of the drive holding the incriminating data (kittens represent illegal material)?

☐ Data

☐ NTFS

☐ Files

☒ Tools

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 24/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. When was the FromPublic directory created?

☐ 2018-03-13 19:36:30

☐ 2018-03-01 5:34:14

☐ 2018-03-13 19:42:36

☒ 2018-03-13 19:33:42

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 28/30

Log into the 172.16.90.100 machine. Analyze (and recover if required) the disk at C:\DFP\Exam_Files\Disks\Disk1.001. Is there any illegal material (kittens represent illegal material)?

- ☐ Stolen documents
- ☐ Illegal book copies
- ☒ Illegal material (kittens)
- ☐ Malicious EXE files

Questions summary



Lab questions



- ☒ Pinned
- ☐ Answered
- ☐ Unanswered

Q: 29/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 in. Which was the first file deleted?

- ☐ CyberAttack-Info.bmp
- ☐ ExamQuestions.docx
- ☒ Quick Stego.Ink
- ☐ netscan.xml

Questions summary



Lab questions



 Pinned

 Answered

 Unanswered

Q: 30/30

Log into the 172.16.90.100 machine. Analyze the C:\DFP\Exam_Files\Disks\Disk3.ad1 image. When was the network share "\shares.dfp.exam\" opened?

☐ 2018-03-13 20:34:34

☐ 2018-03-13 20:31:34

☒ 2018-03-01 5:45:28

☐ 2017-03-13 20:34:12